Proiect SCIA

Gheorghe Alexandru

Grupa:2132

1.Tematica proiectului

a)Etajul 1: Amplificator de instrumentatie V-V cu 3 AO cu intrari diferentiale

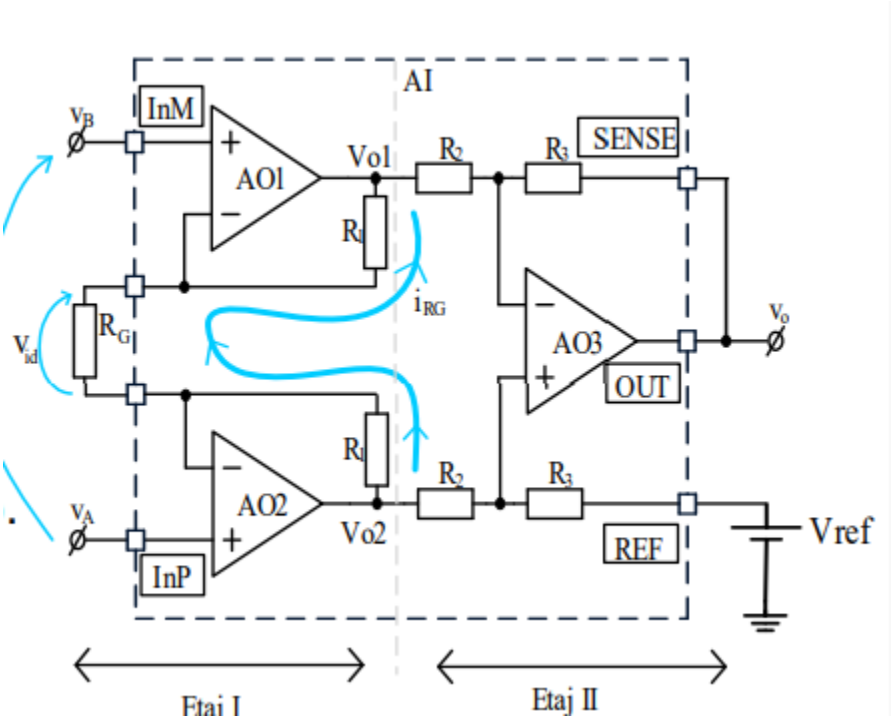
b)Etajul 2:Filtru trece jos cu 3 AO V-V KHN

c)Etajul 3:PGA , switch-uri in calea de semnal, conexiune serie

d)Etajul 4:Redresor detector de varf.

2.Dimensionarea Etajelor

a)Etajul1:



AO1 = Reactie negativa => $V^+ = V^- = V_B$;

AO2 = Reactie negative => $V^+ = V^- = V_A$;

$V_{RG} = V_A - V_B = V_{ID}$; $i_{RG} = V_A - V_B / R_G$;

$V_{O2} - V_{01} = (R1 + RG + R1)*i_{RG}$;

$A_{d1} = V_{02} - V_{01}/V_A - V_B = (1 + 2R_G/R_G)$;

$V_{out} = (V_{02} - V_{01})\ (R_3/R_2) + V_{REF}$;

$A_{d2} = R_3/R_2$;

$G = A_d = (1+2R_1/R_G)* R_3/R_2 = 4$

Alegem : $R_3 = R_2$

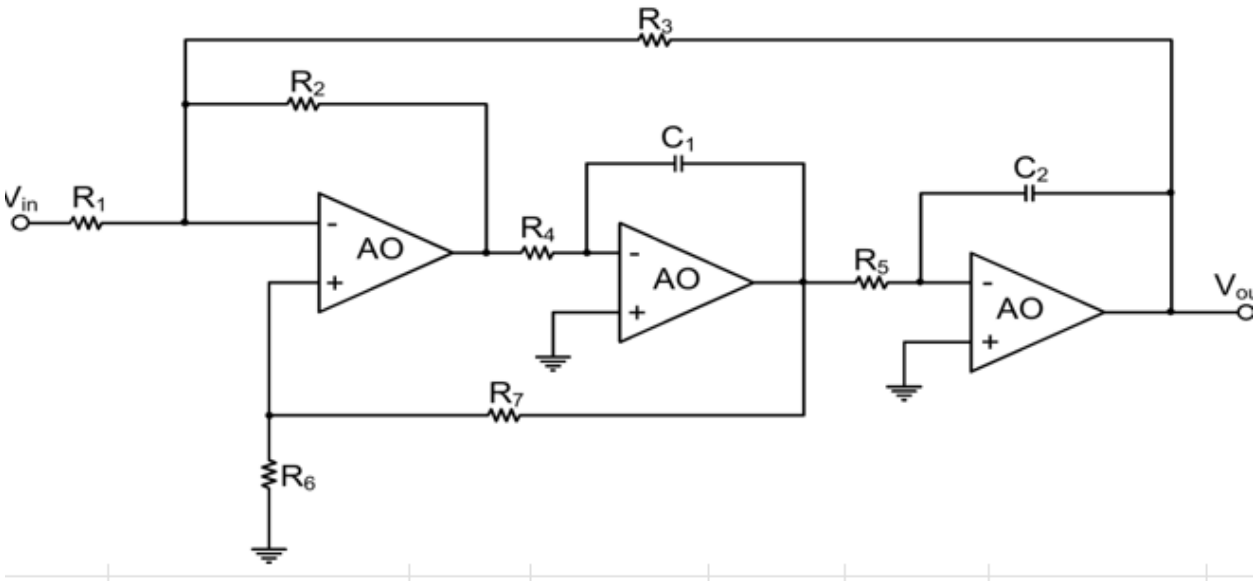
$1+2R1/R_G = 4$

$R_G = 2R_1/4$

Alegem R_1=10K = > R_G = 6.67K=> 6.8K val standard

$R_1 = R_2 = R_3$ =10K

b)Etajul 2:



Castig linear in banda de trecere: 1

R intrare minim: 750

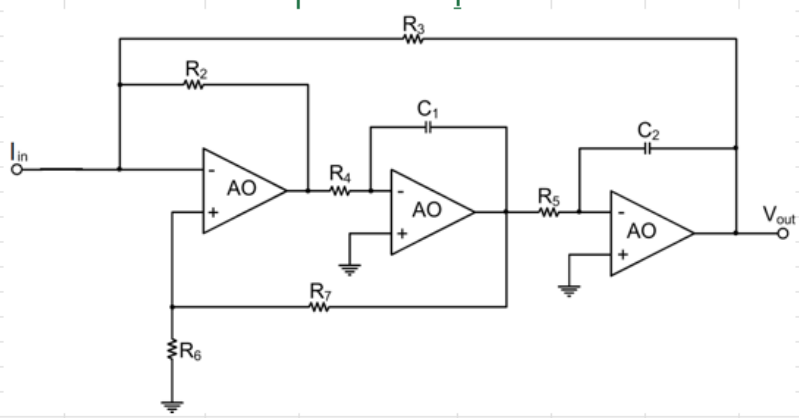
Banda:4kHZ

Q = 0.707

Alegem R: 2k

f0 [Hz]	w0[rad/sec]	Q	H0[V/V]
4.00E+03	2.51E+04	7.07E-01	1.00E+00
	varianta 1		
	C1[F]	set C2[F]	set R[ohm]
	2.11E-08	1.88E-08	2.00E+03

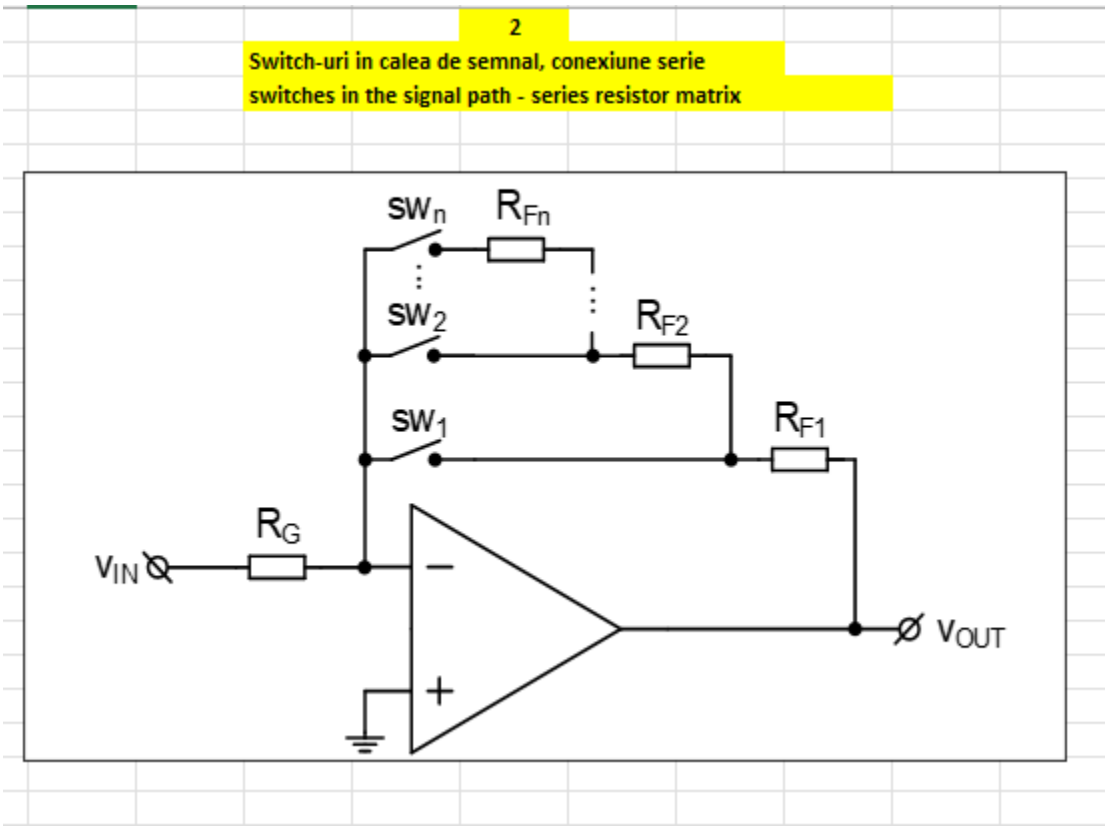
$$C_1 = \frac{3Q}{2\omega_0 R}; C_2 = \frac{4C_1}{9Q^2} = \frac{2}{3Q\omega_0 R}$$



C1 = 21.1nF

C2 = 18.8nF

c)Etajul 3:



Castig minin [dB]=1

Casig maxim[dB]=7

Rezolutie[dB] = 2

Castig: 1db , 3db, 5db, 7db

$$A_v = 10^{\frac{A_v(\mathrm{dB})}{20}}$$

$$A_v = 10^{1/20} \approx \boxed{1.122}$$

$$A_v = 10^{3/20} \approx \boxed{1.413}$$

$$A_v = 10^{5/20} \approx \boxed{1.778}$$

$$A_v = 10^{7/20} \approx \boxed{2.239}$$

Alegem R_G = 10k :

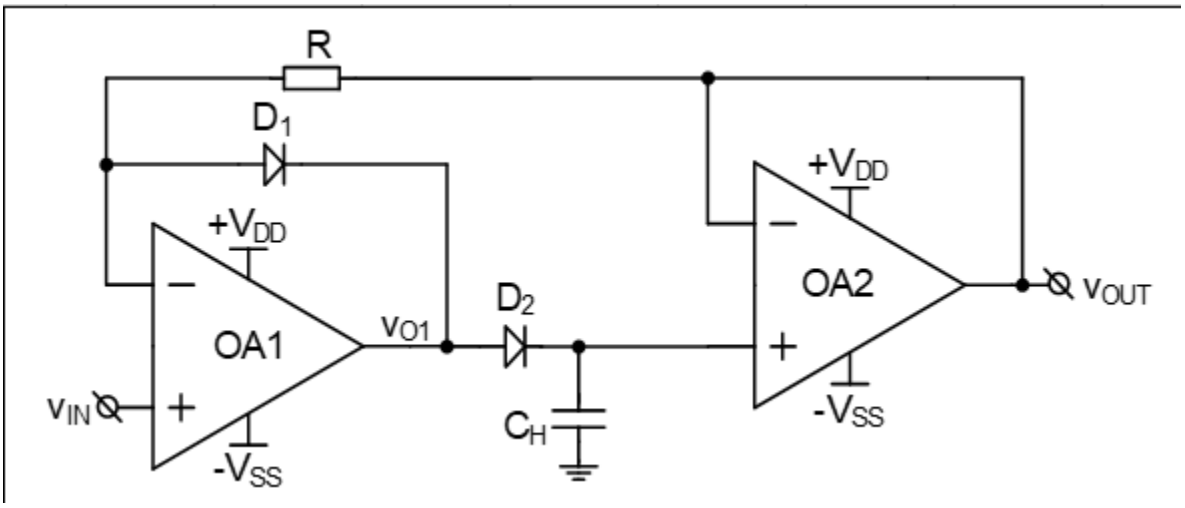
R_{F1}/R_G=1.122 => 11.2k

$R_{F1}+R_{F2}/R_G$=1.413=>14.1K

$R_{F1}+R_{F2}+R_{F3}/R_G$ = 1.778 => 17.8k

$R_{F1}+R_{F2}+R_{F3}+R_{F4}/R_{G\,=\,2.239}$ = > 22.4k

d)Etaj 4:



Consideram D_1,D_2 OFF

$V_{IN}>0 \Rightarrow V_{IN} \rightarrow V_{DD} \Rightarrow D_1$ OFF , D_2 ON , $V_{CH=0}$

OA2 = Reactie negativa => $V^+=V^- = V_{in}$

OA1= Reactie negative prin OA2 => $V^+=V^-= V_{in}$

$V_{OA1}=V_{in} \Rightarrow V_{CH} \rightarrow +V_{in_max}$

$V_{OUT}= V_{in_max}$

$V_{IN}<0 \Rightarrow V_{IN} \rightarrow -V_{SS} \Rightarrow D_1$ ON, D_2 OFF, $V_{CH} = V_{DD}$ (din starea anterioara)

V_{OA2} = Reactie negativa =>$V^+=V^-= V_{CH}= V_{in_max}$ =>

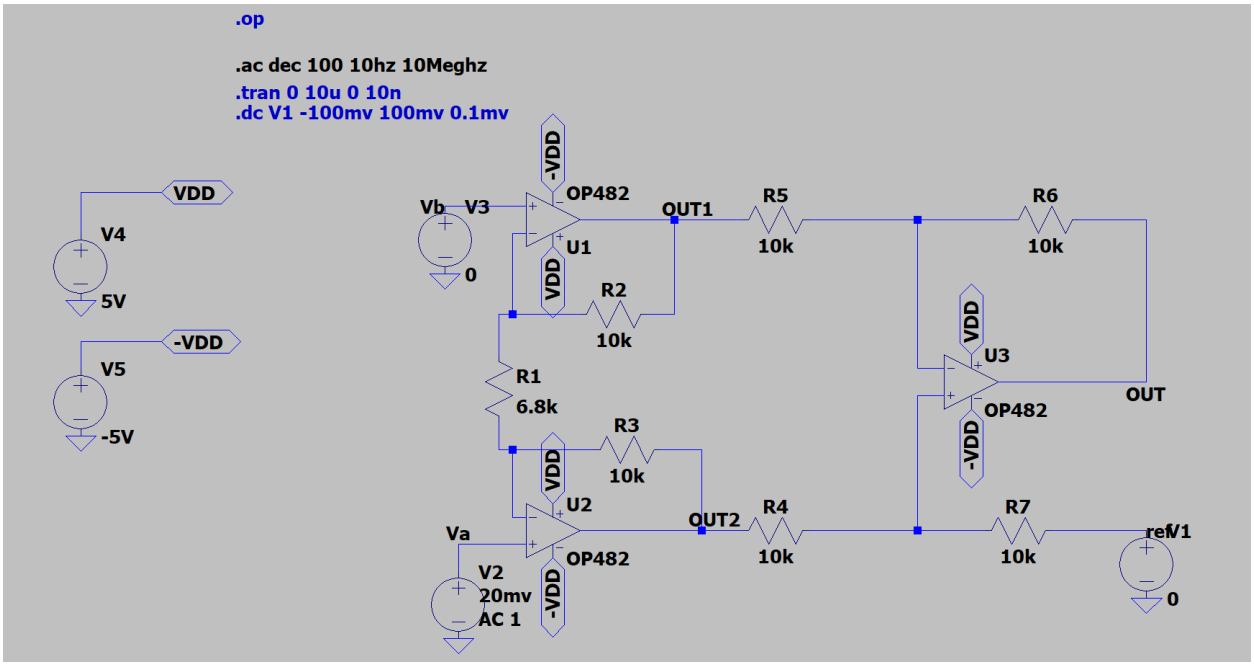
=>$V_{OUT}= V_{in_max}$;

$V_{OUT} = +V_{in_max}$, $V_{IN} > 0$

$V_{OUT} = +V_{in_max}$, $V_{IN} < 0$

3.

a)Etaj 1:



PSF:

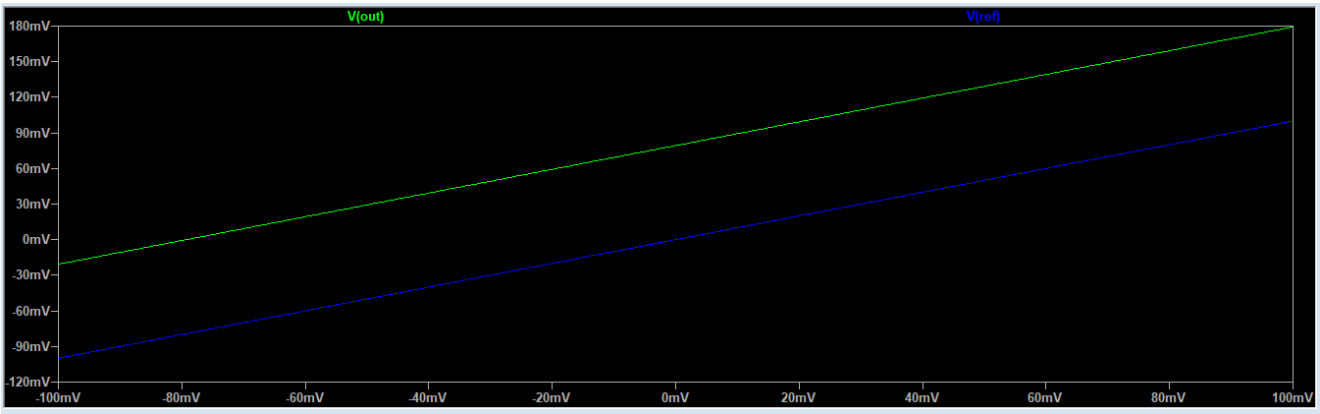
V(-vdd):	-5	voltage
V(n001):	-0.000879829	voltage
V(n002):	-0.000586568	voltage
V(n003):	-0.000293288	voltage
V(n004):	-0.000586551	voltage
V(out):	-0.0011731	voltage
V(out1):	-0.000586663	voltage
V(out2):	-0.0005866	voltage
V(ref):	0	voltage
V(va):	0	voltage
V(vb):	0	voltage
V(vdd):	5	voltage

Iesirile U1,U2 =/= +-5V

Amplificatoarele nu functioneaza in regim de saturatie

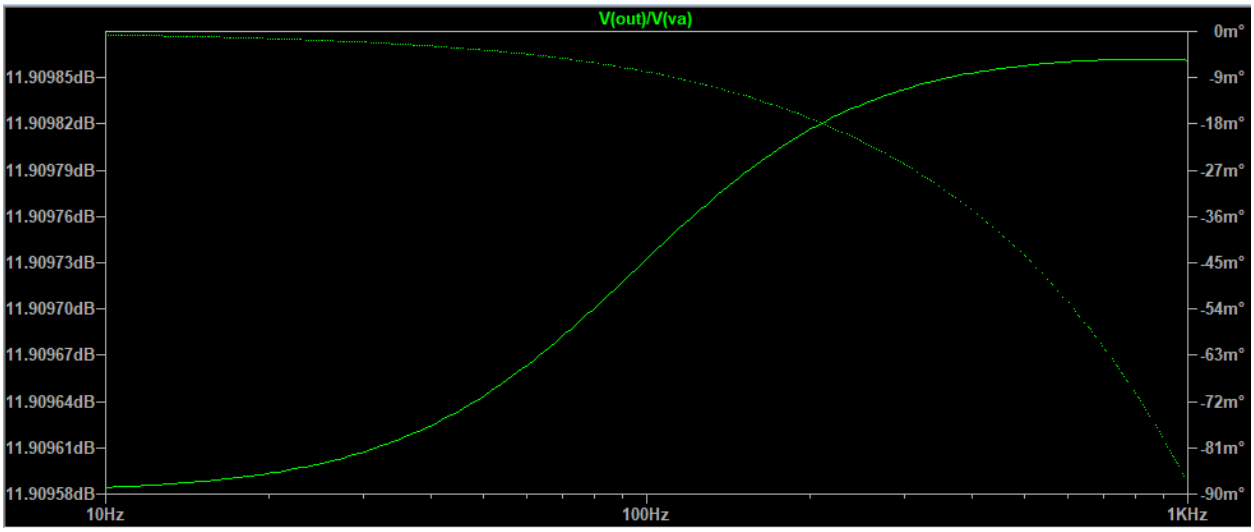
$U_{out} = G*(V_A - V_B)$

Compensare DC:



V(out) se deplaseaza linear cu tensiunea de referinta.

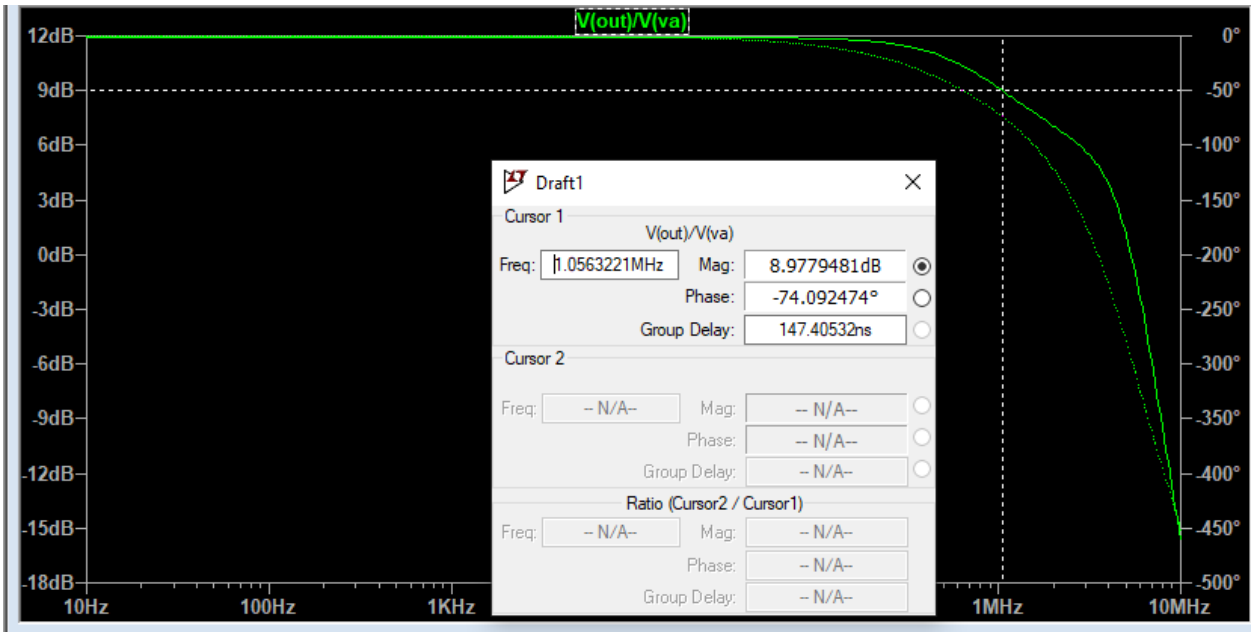
AC:



$V_{out}/V_{in} = 11.9dB$

G=4

20 $\log_{10}(4)$=12dB



Banda = 1.056Mhz

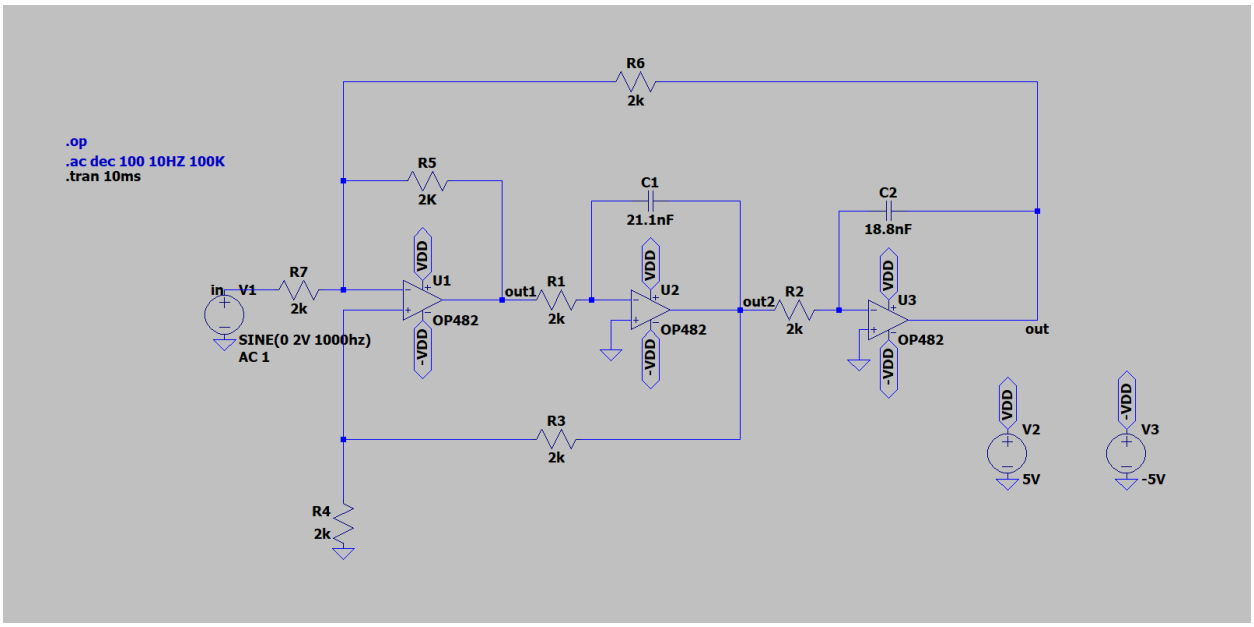
Banda filtru = 4000hz

Banda > Banda filtru

Transient

SR =0.2V/us

b)Etaj 2 :



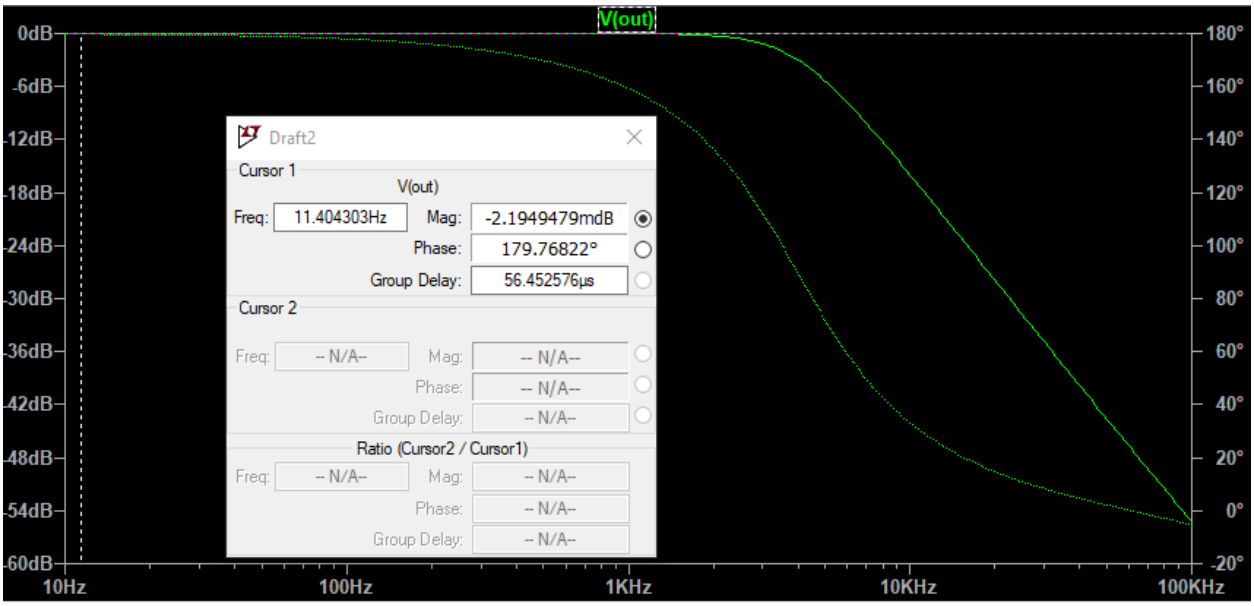
DCOP:

V(out):	0.000812817	voltage
V(out1):	0.000232235	voltage
V(out2):	0.000232168	voltage

$V_{OUT} = 0$

$V_{OUT1,2}$ nu se afla la saturatie

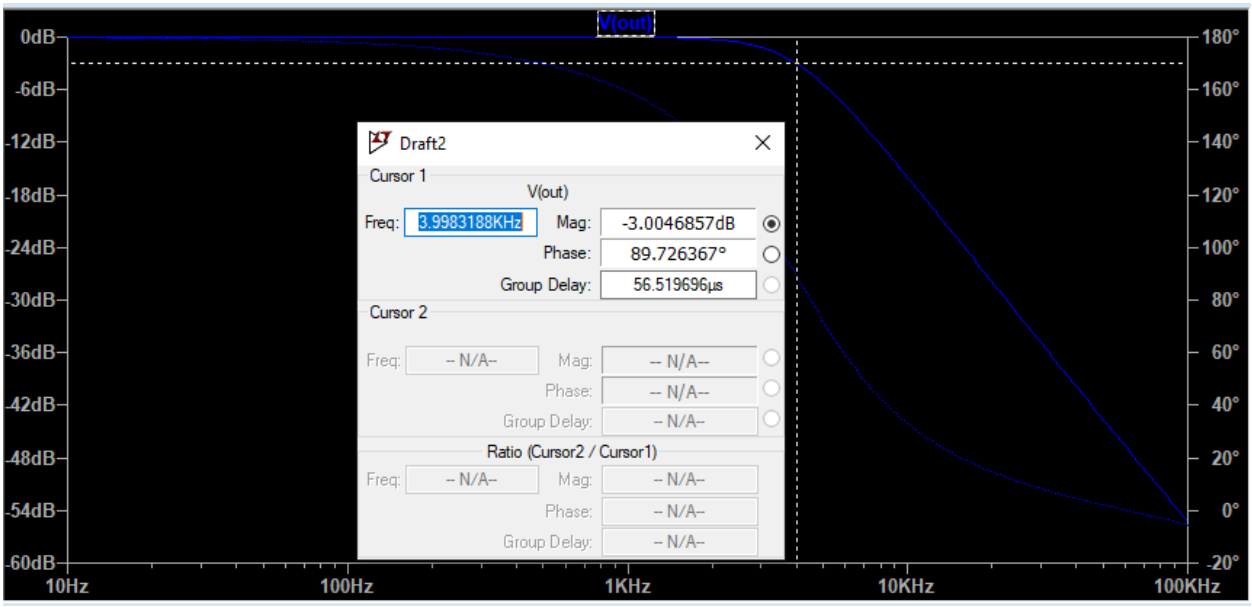
AC:



Casigul in banda de trecere este -2mdB=>

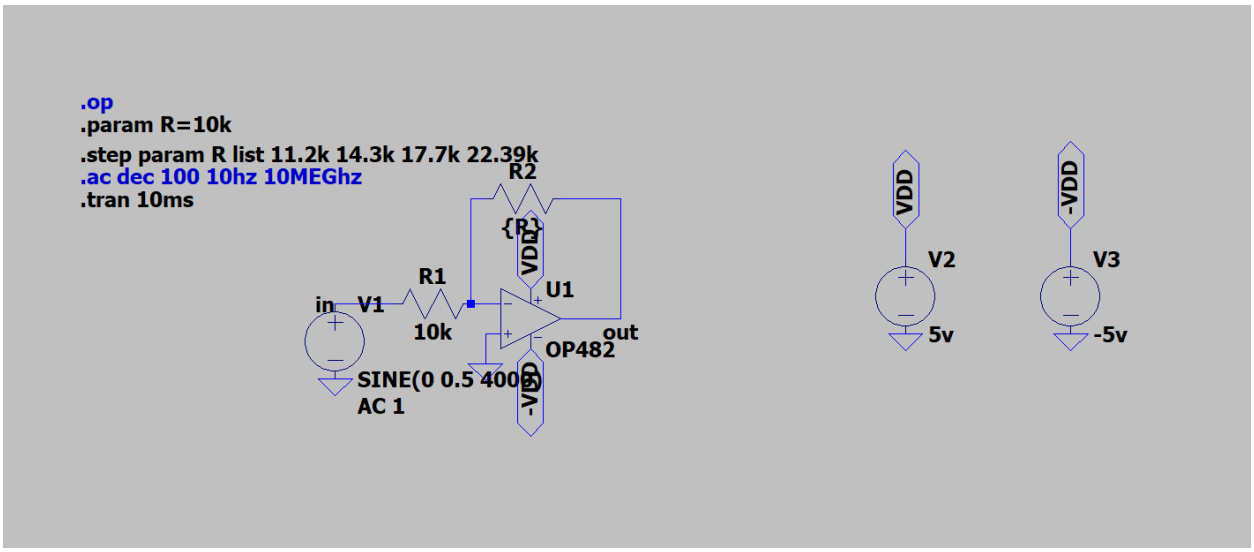
$$V = V_0 \cdot 10^{\frac{dB}{20}} = 1 \cdot 10^{0.002/20} = 10^{0.0001} \approx 1.0002303 \text{ V}$$

Castigul in banda de trecere este aproximativ 1



Castigul la 3.99Khz = -3dB =>Banda = 4Khz

c)Etaj 3:

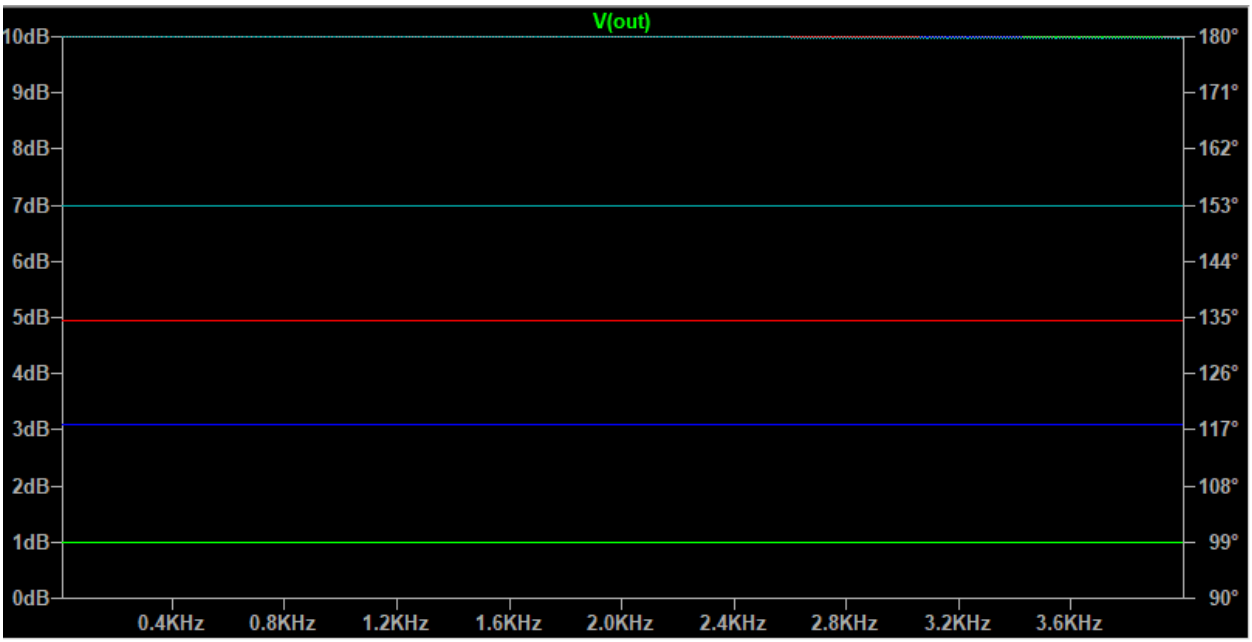


DCOP:

V(in):	0	voltage
V(n001):	0.000232233	voltage
V(out):	0.000464485	voltage

Amplificatorul functioneaza in regim de nesaturatie.

AC:

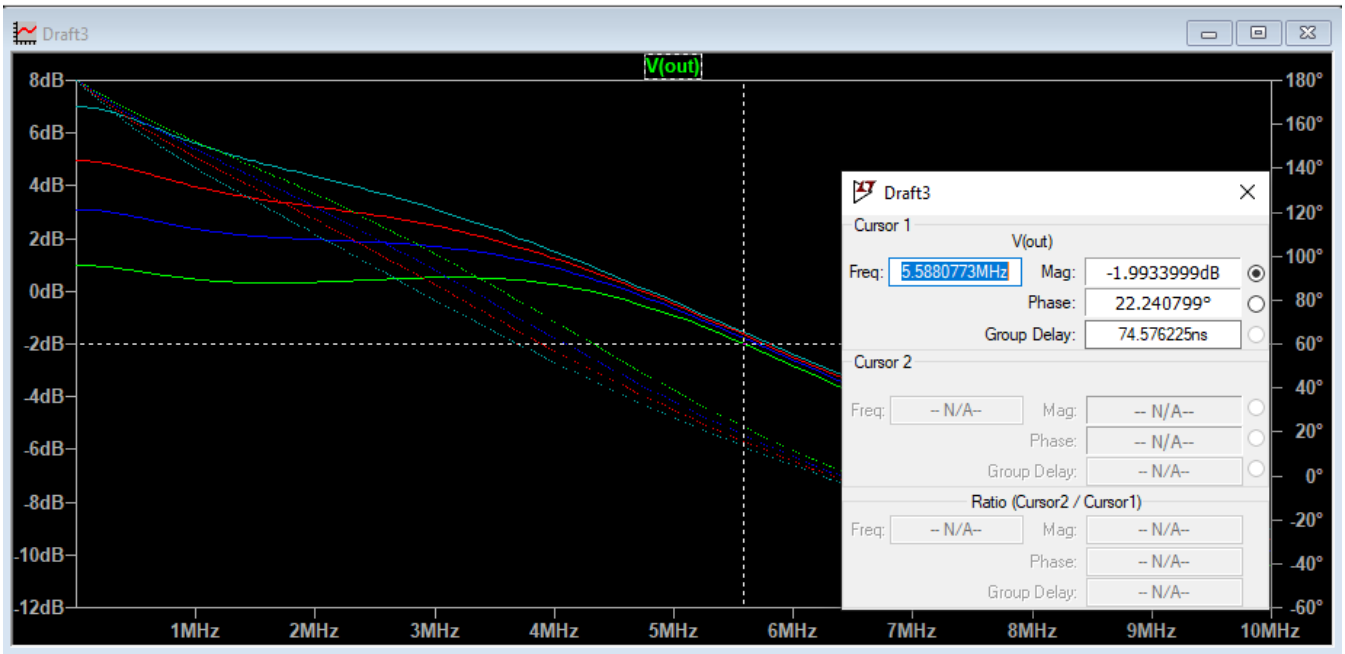


Castig minin [dB]=1

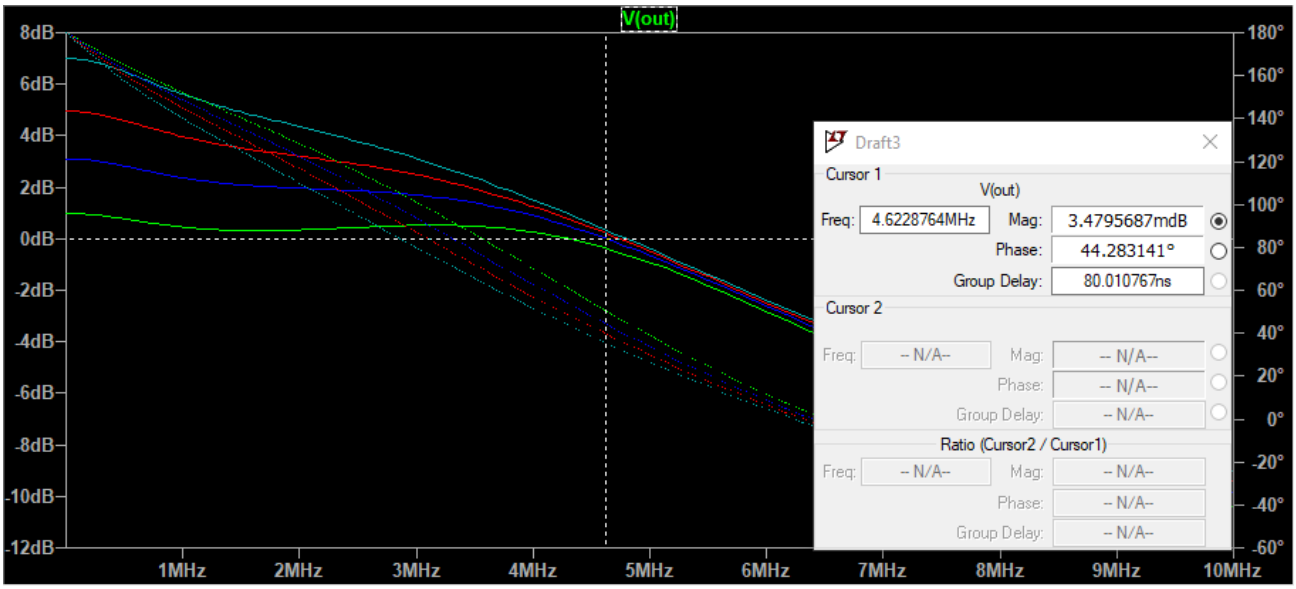
Casig maxim[dB]=7

Rezolutie[dB] = 2

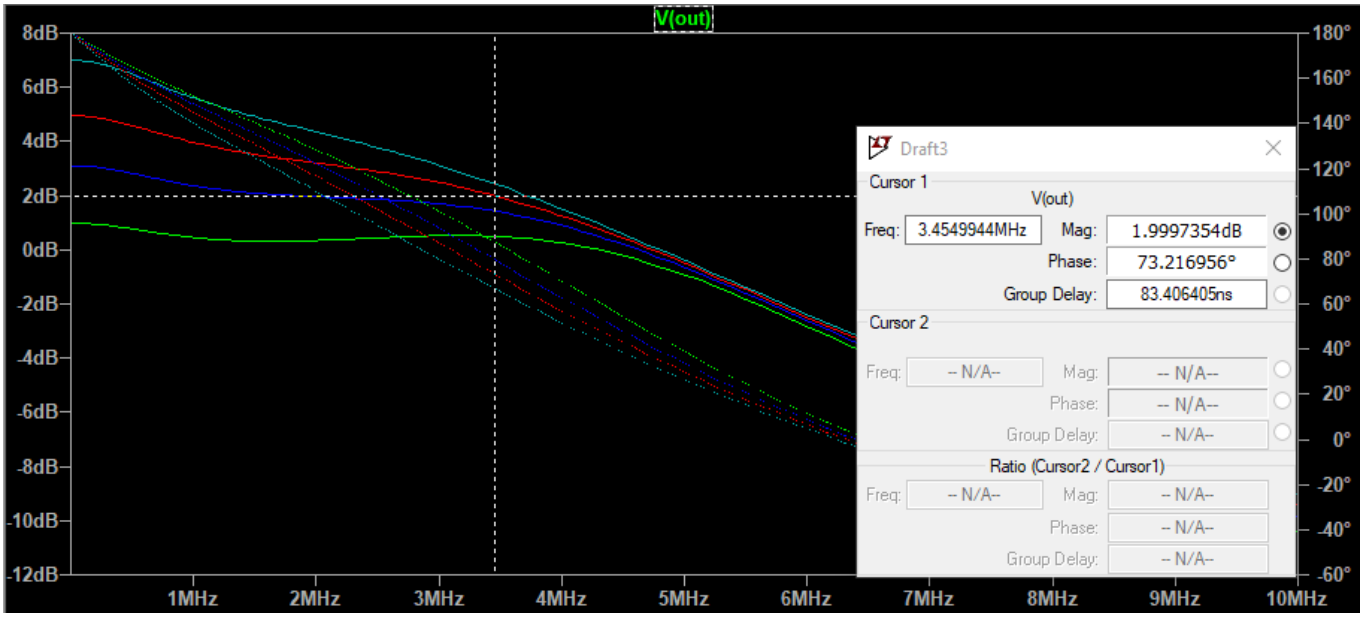
Castig: 1db , 3db, 5db, 7db



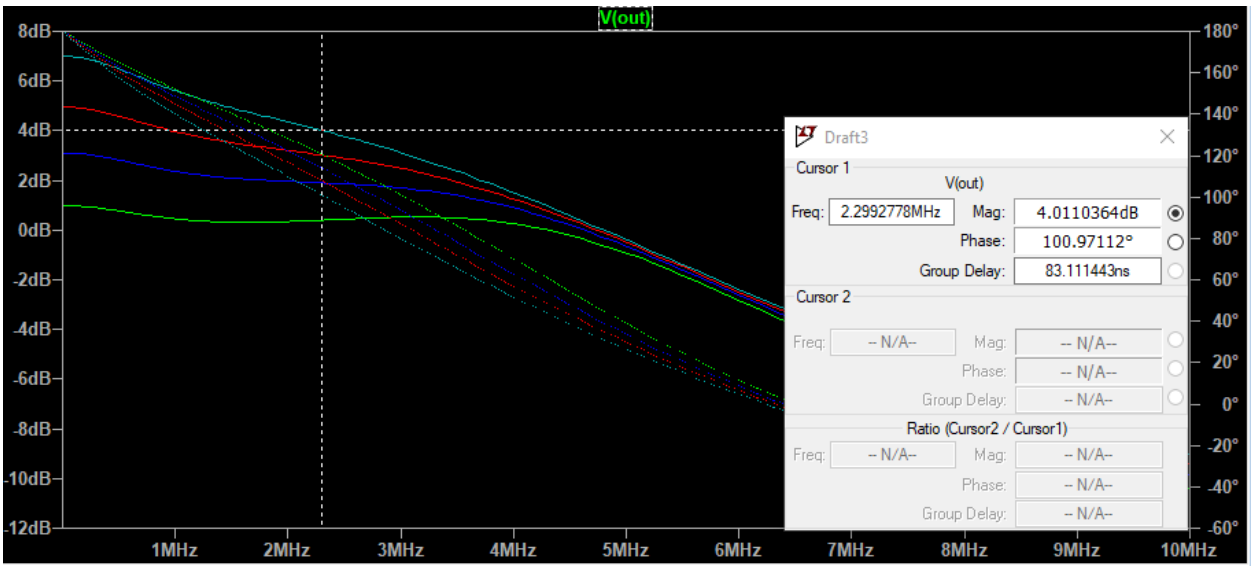
Castig= 1db=> Banda=5.6Mhz > banda filtru=4khz



Castig= 3db => Banda=4.6Mhz > banda filtru = 4khz

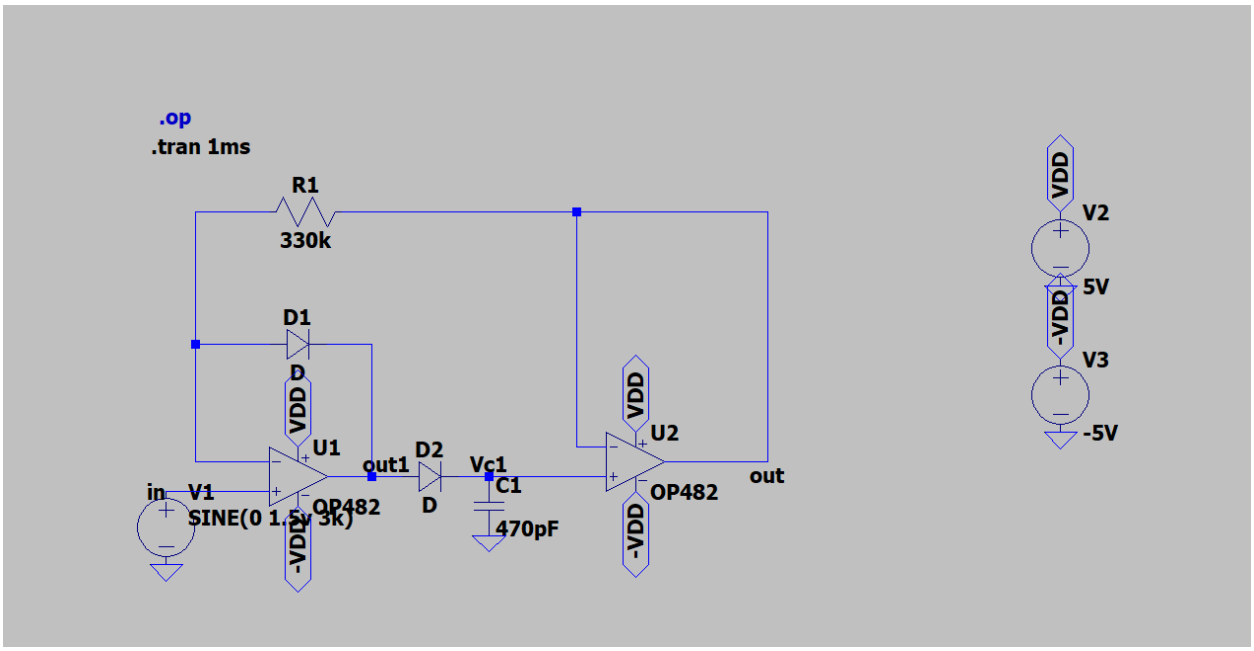


Castig = 5db => Banda = 3.4Mhz > Banda filtru = 4khz



Castig 7db => Banda = 2.29Mhz > Banda filtru = 4khz

d)Etaj 4:



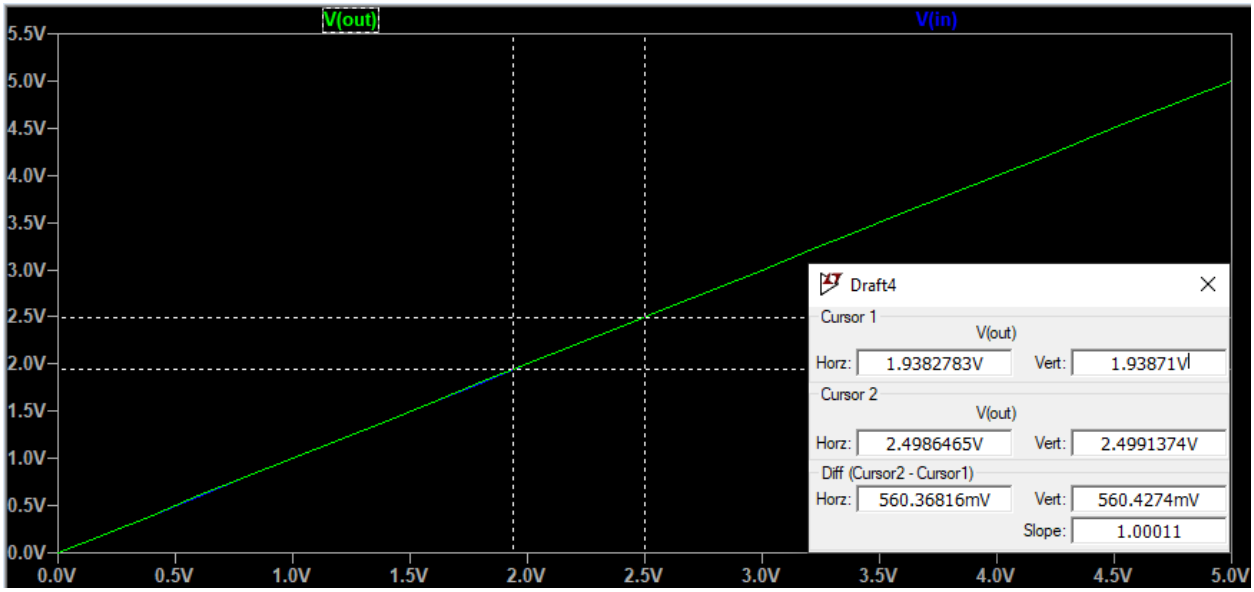
DCOP:

V(in):	0	voltage
V(n001):	0.000228831	voltage
V(n002):	-3.39883e-06	voltage
V(out):	0.000228848	voltage
V(out1):	0.108139	voltage
V(vdd):	15	voltage

$V_{in}=0=V_{out}=V_{out1,2}$

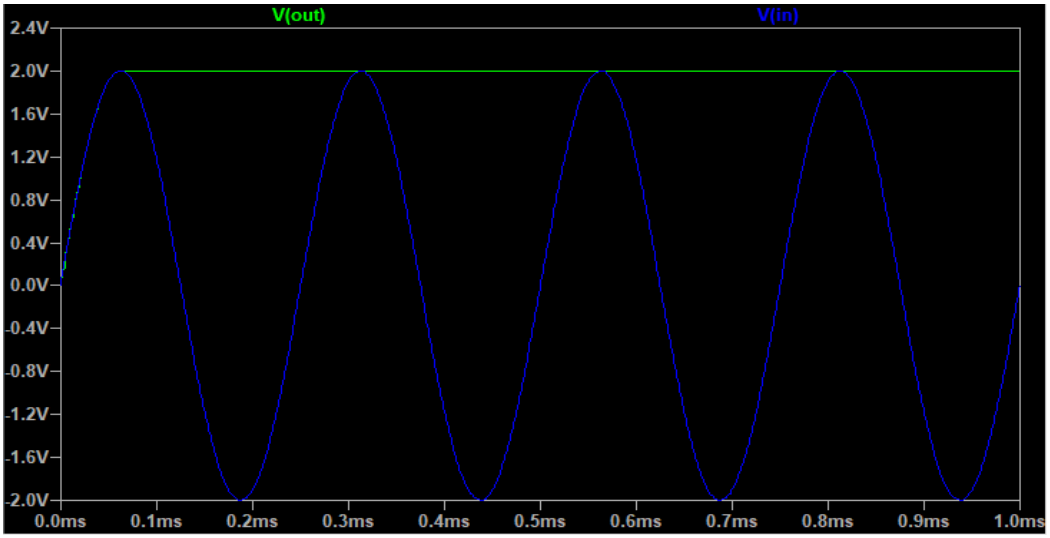
Amplificatoarele nu se afla in regim de saturatie.

DC



$V_{out} = V_{in}$ => Castigul liniar = 1

Transient:



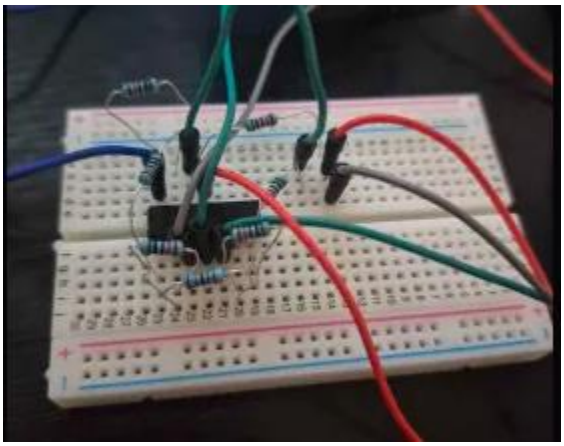
$V_{OUT} = +V_{in_max}$, $V_{IN} > 0$

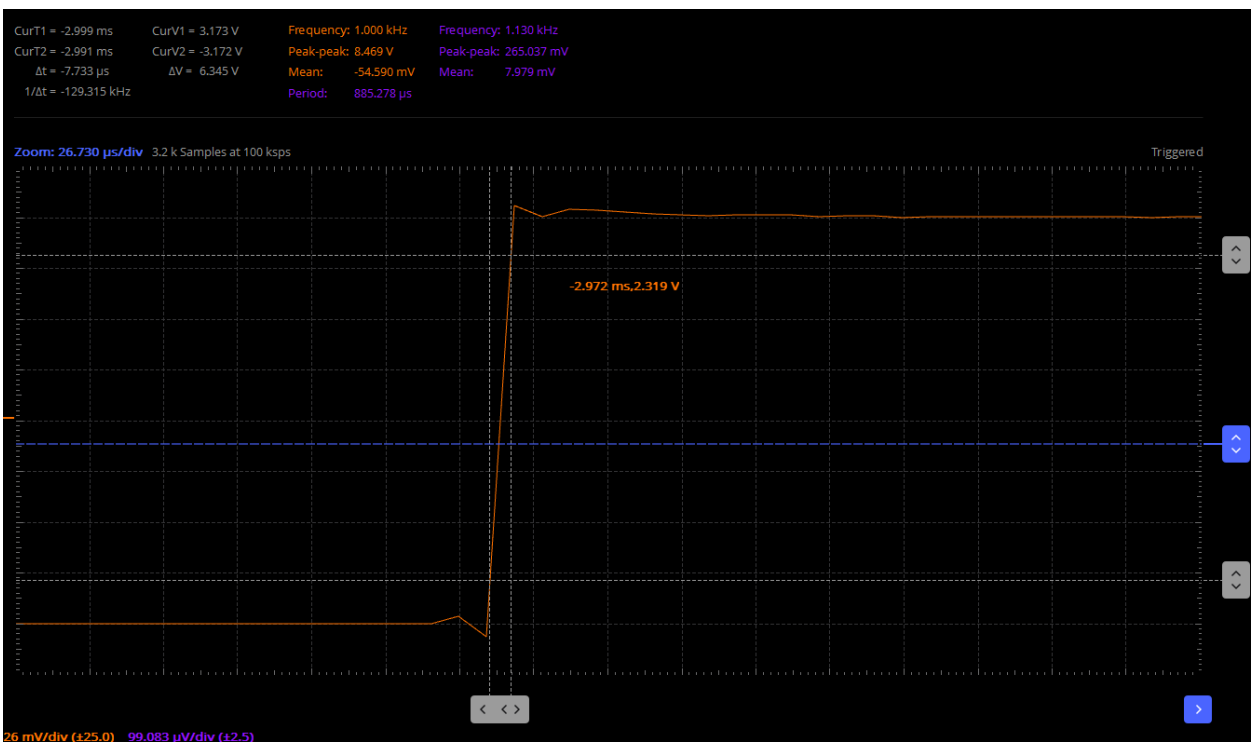
$V_{OUT} = +V_{in_max}$, $V_{IN} < 0$

Dupa prima alternanta pozitiva $V_{out} = V_{in_max}$

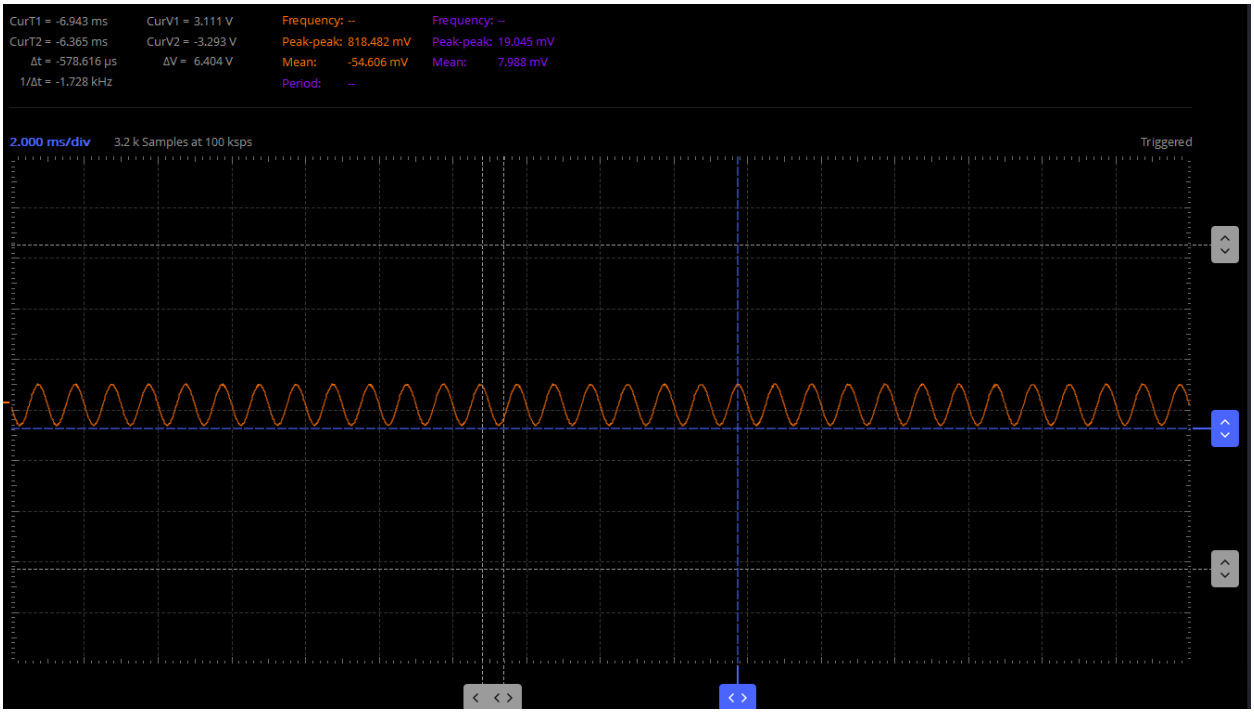
4)Masuratori

Etaj 1:



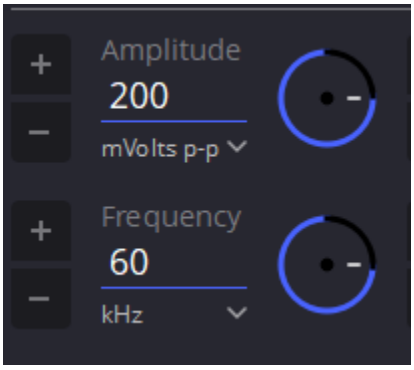


SR=0.81V/us



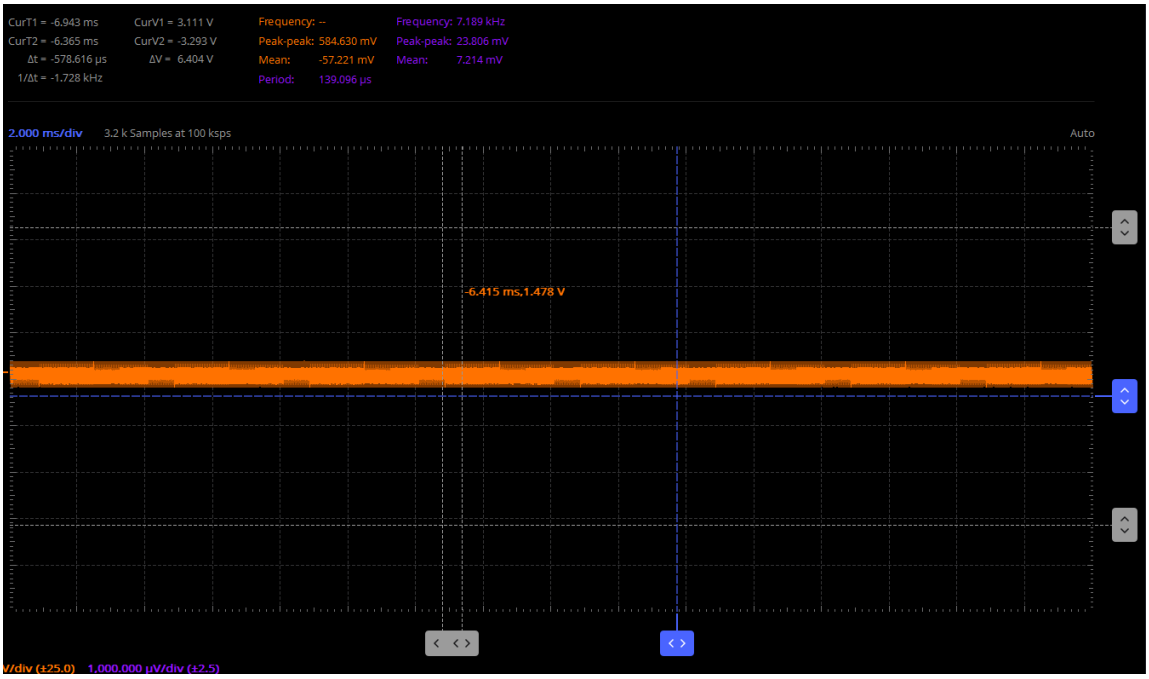
Setam un semnal sinusoidal $V_{in\ peak\ to\ peak}$=200mv =>

V_{out}=818mv => $V_{out\ la\ -3db}$=578mv

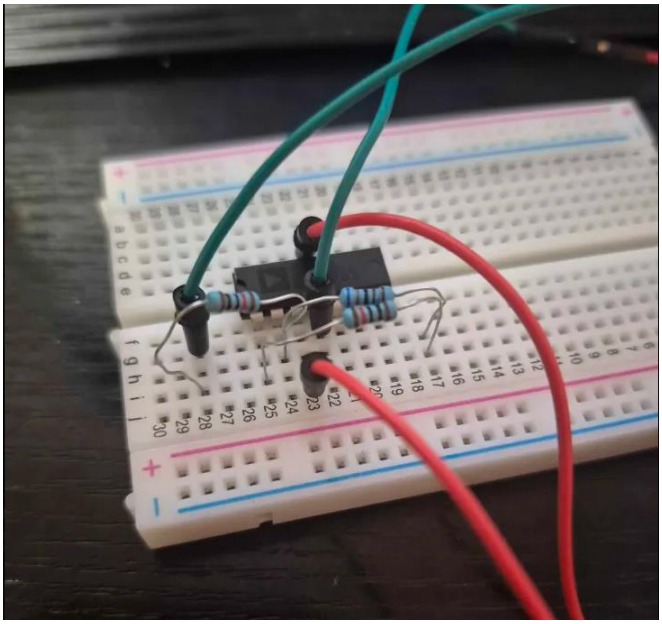


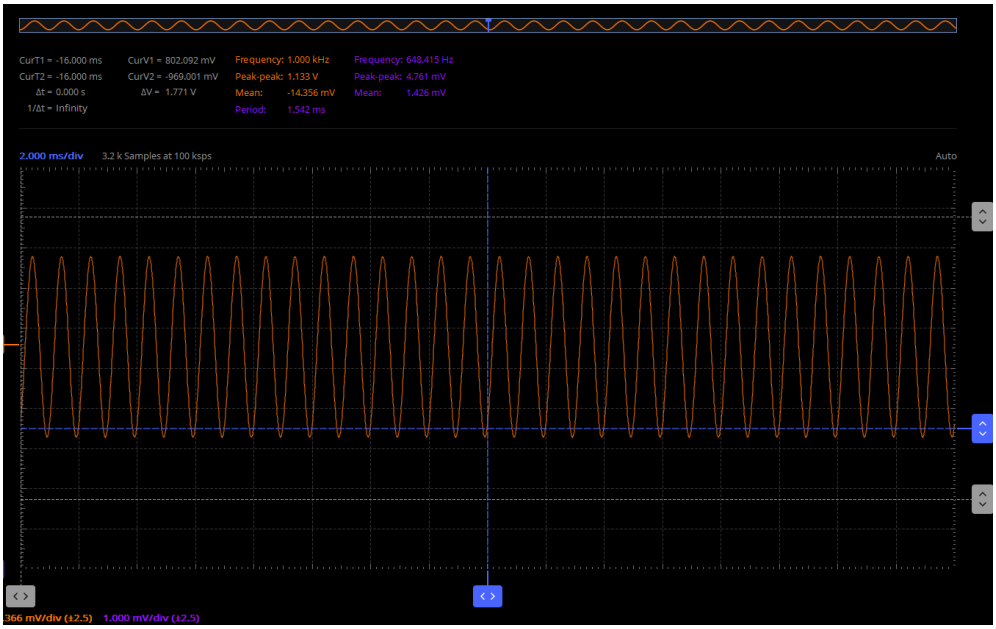
La o frecventa de 60 kHz avem $V_{out\ peak\ to\ peak}$ =584 =>

Banda aproximativ egal cu 60kHz > Banda filtru = 4kHz


CurT1 = -6.943 ms
CurT2 = -6.365 ms
Δt = -578.616 µs
1/Δt = -1.728 kHz
CurV1 = 3.111 V
CurV2 = -3.293 V
ΔV = 6.404 V
Frequency: --
Peak-peak: 584.630 mV
Mean: -57.221 mV
Period: 139.096 µs
Frequency: 7.189 kHz
Peak-peak: 23.806 mV
Mean: 7.214 mV
2.000 ms/div
3.2 k Samples at 100 ksps
Auto
-6.415 ms,1.478 V
V/div (±25.0)
1,000.000 µV/div (±2.5)

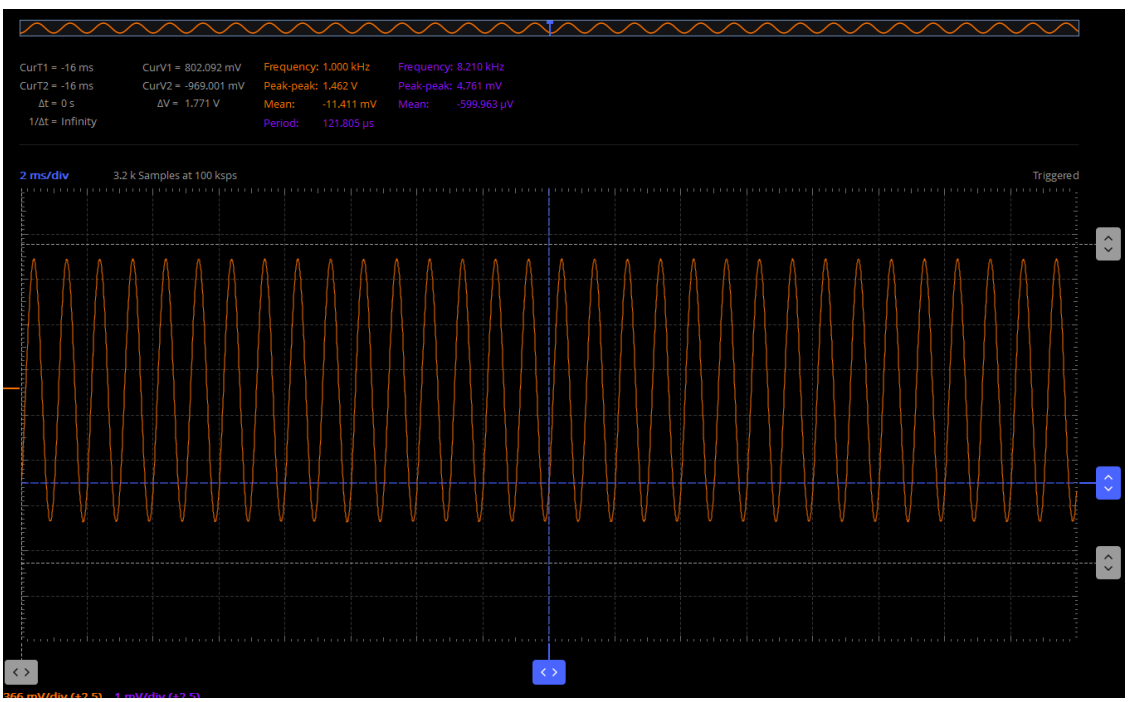
Etaj 3:





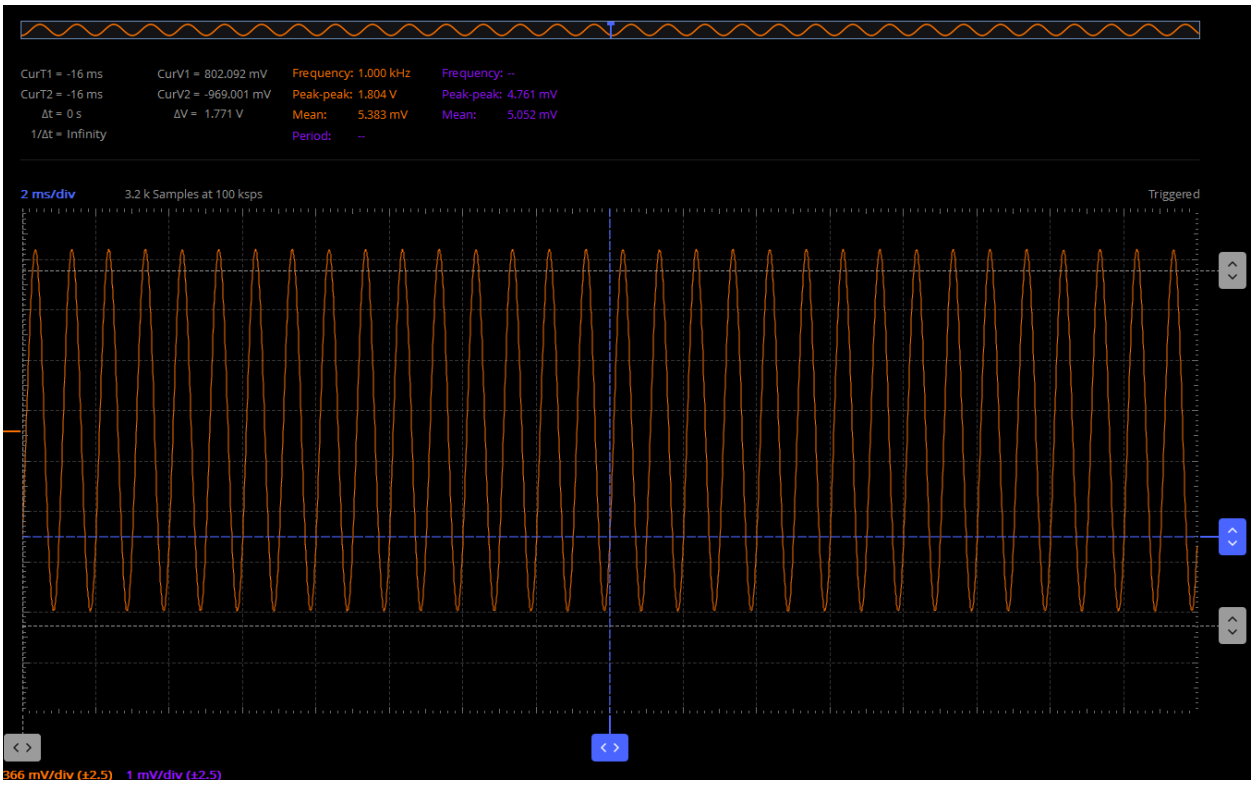
Semnal de intrare $1V_{pp}$ =>V_{out} = 1.133V

Amplificare= 1.133 = 0.9db



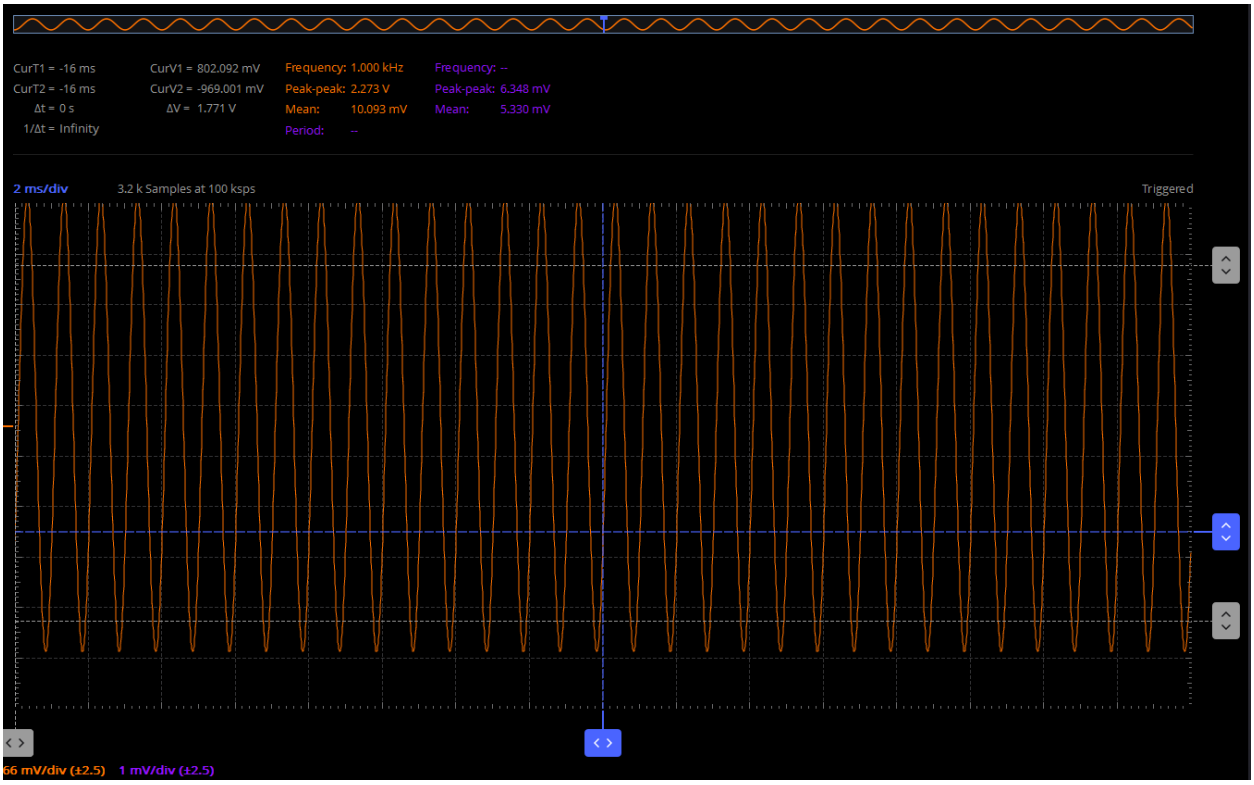
Semnal de intrare $1V_{pp}$ =>V_{out} =1.46V

Amplificare = 1.46 = 3.3db



Semnal de intrare $1V_{pp}$ =>V_{out} =1.8V

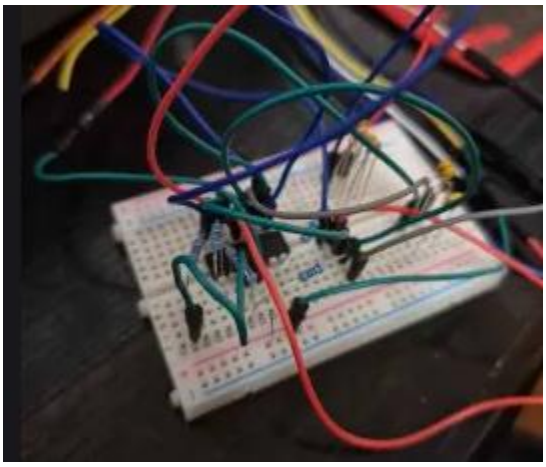
Amplificare = 1.8= 5.1db

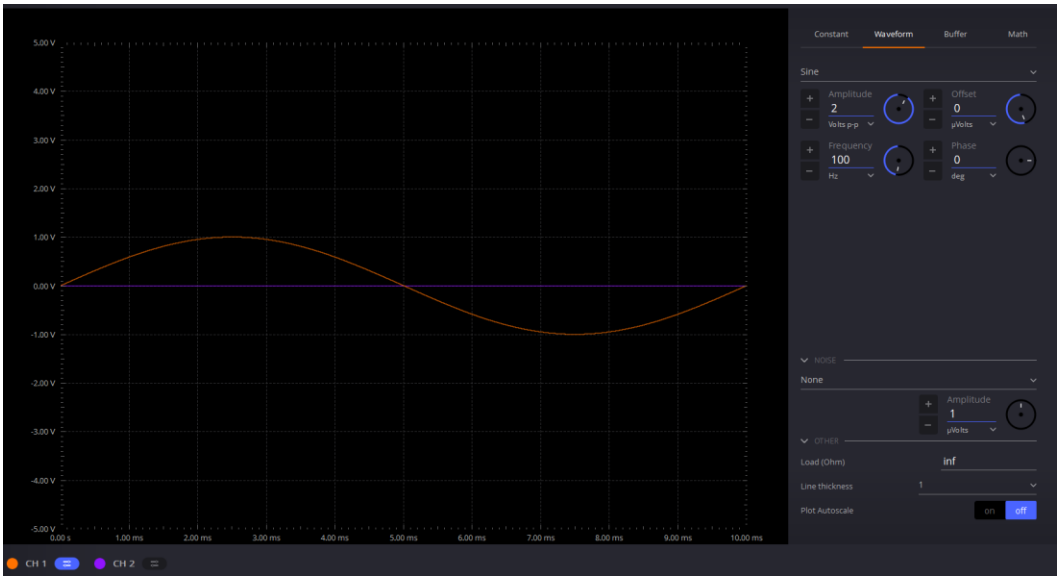


Semnal de intrare $1V_{pp}$ =>V_{out} =2.27V

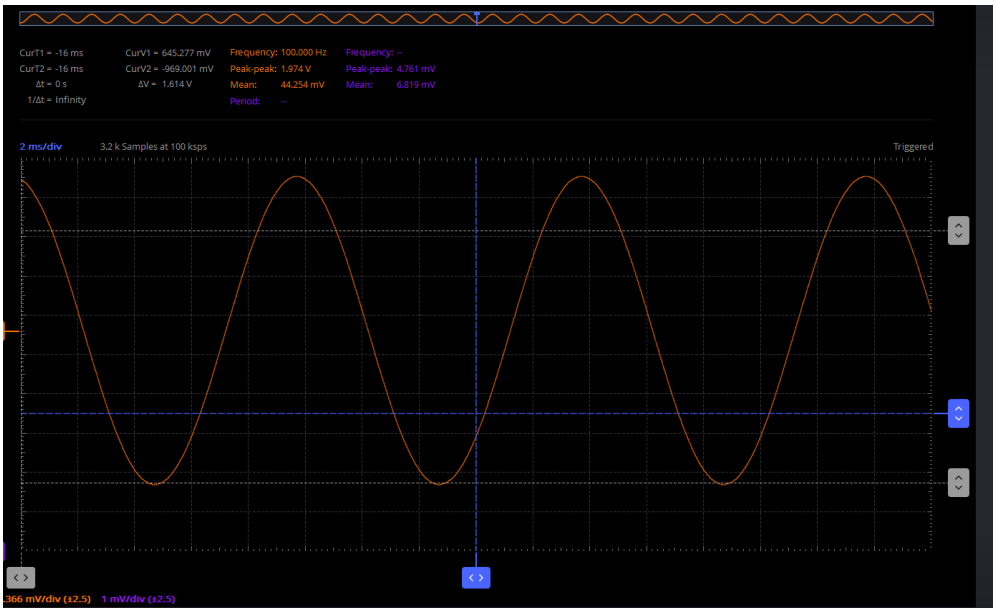
Amplificare = 2.27 = 7.1db

Etaj 2:



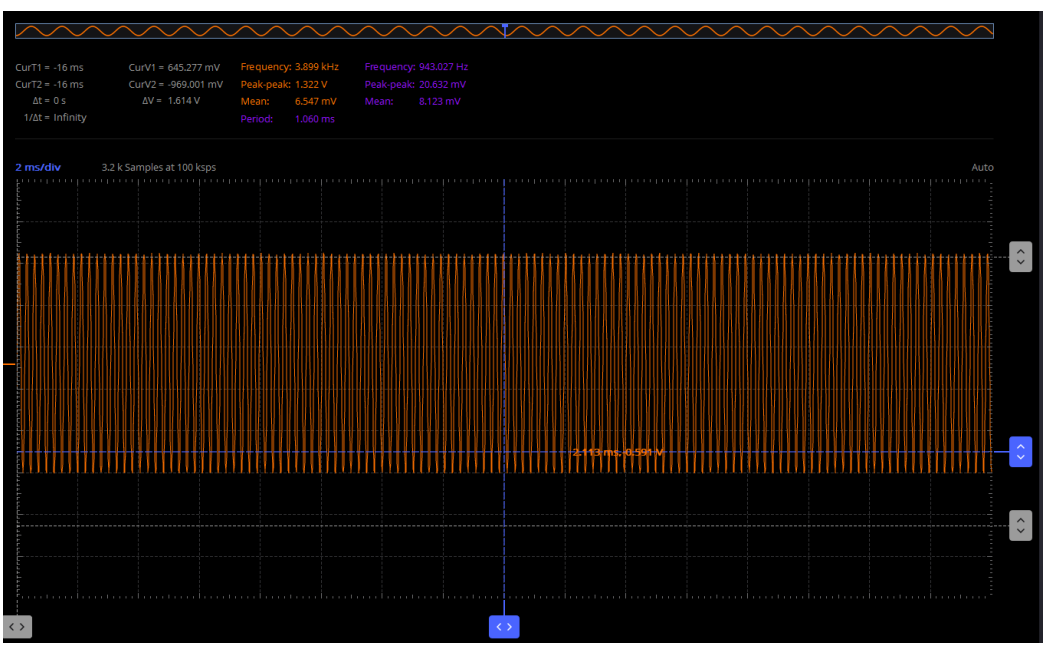


Generam un semnal de $2V_{\text{peak to peak}}$ cu frecventa de 100Hz



V_{out} = 1.974V, V_{in}=2V => Castigul in banda de trecere este aproximativ 1.

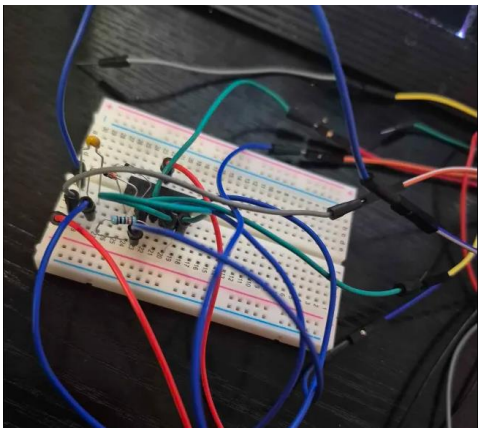
V_{out} = 1.974=> V_{out} la -3db = 1.3V

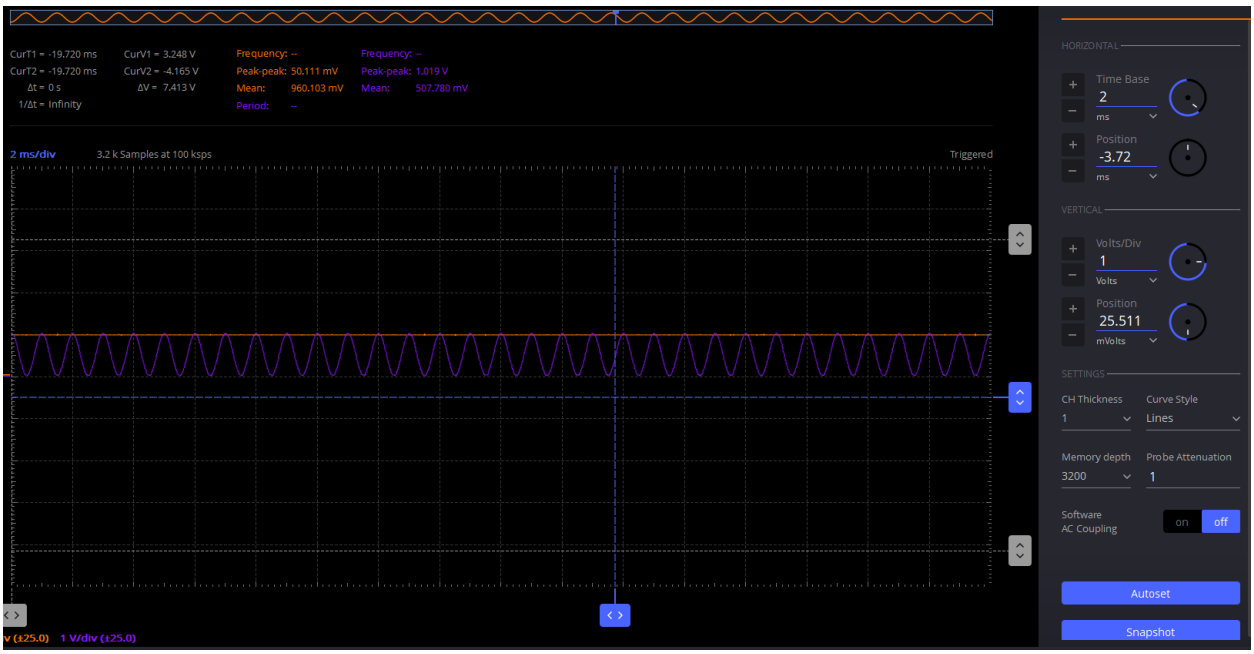


La frecventa 3.9kHz circuitul are o amplificare de -3db

⇨ Banda = 3.9kHz

Etaj 4 :





Functia circuitului:

$V_{OUT} = +V_{in_max}$, $V_{IN} > 0$

$V_{OUT} = +V_{in_max}$, $V_{IN} < 0$

5)

a)Etaj 1:

G=4=> Castig in decibel = 12db

Castigul din simulare = 11.9 db

Diferenta este datorata folosirii valorilor standardizate de rezistente $(1+2*R_1/R_g)=(1+2*10k/6.8k) = 3.94$

b)Etaj 2:

Castigul linear in banda de trecere din specificatii = 1

Din simulator am obtinut = 1.0002303

Banda filtrului ceruta in specificatii = 4khz

Banda obtinuta in circuit = 3.99Khz

c)Etaj 3:

Castiguri PGA cerute in specificatii: 1dB,3dB,5dB,7dB

Castiguri PGA obtinute:983mdB,3.1dB,4.9dB,6.9dB

d)Etaj 4:

Castigul linear din specificatii = 1

Castigul obtinut= V_{out}/V_{in}= 1.09

Etaj 1:

Simulare	Masurare
Banda = 1Mhz	Banda = 60kHz
SR=0.2V/us	0.81V/us

Diferenta in banda deoarece banda in circuitul fizic din cauza efectelor reale care nu sunt surprinse in simulare: capacitate parasite,indunctanta firelor,capacitatea sondelor (10-15pF).

Teoretic fiecare AO are o banda limitata cu 3 AO banda totala scade in simulare etajele sunt decuplate.

Slew rate-ul este mai mic in simulare deoarece modelul din LTSpice este “worst case” si respecta valoarea minim garantata si limiteaza curentii interni mai sever, in realitate circuitele integrate sunt mai bune decat minimul din datasheet.

Etaj 2:

Simulare	Masurari
Castigul in banda de trecere: 1.0002	Castigul in banda de trecere: 0.98
Banda: 3.99kHz	Banda: 3.9kHz

Rezultatele obtinute din masurari sunt aproximativ identice cu cele din simulari

Etaj3:

Simulare	Masurari
Castig 1: 0.98db	Castig 1: 0.9db
Castig 2: 3.1db	Castig 2: 3.3db
Castig 3: 4.9db	Castig 3: 5.1db
Castig 4: 6.9db	Castig 4: 7.1db

Valorile din masurari sunt aproximativ egale celor din simulare